ONR FY2027 YIP: second technical review

Trustworthy Collective Reasoning and Evaluation for AI Agents

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Version and scope. I read the revised technical narrative in full and incorporated the subsequent changes through repository main commit `6682ee9af5c0a3615a1f5a45ab309fa5e292b5bc`, PDF Git blob `a9d02f5b626da8b319993e5fa2435ed8dc5ccd7e`. Sections 1-6 now occupy numbered pages **1-24** (PDF physical pages 3-26). Because the latest request asks for the significantly revised proposal, I include all 24 technical pages; the earlier instruction to ignore the CV and other back matter still applies. Page locations below are current printed technical pages. Equations were read from the LaTeX. This is a simulated panel assessment from three perspectives, not independent external panel reports.

Review threshold. This review follows your instruction to prioritize material problems. C3-C10 are the eight decision-relevant issues. Minor assumptions and edge cases appear only in the math audit, clearly separated from that list. Corrected objections from v1 are closed rather than repeated. The proposed fixes retain the central question, three RGs, five tasks and section order. Each substantive addition names the passage it replaces or a corresponding cut. No new predeadline experiments are requested beyond the five pilots already planned.

1. Verdict and what changed

C1. Borderline scientifically; not submission-ready at 24 technical pages

Location: Sections 1.3, 3.1-3.7 and 6, pp. 3-24; eqs. (1)-(6), Propositions 1-4.

Technical merit: Good. The revised proposal connects its methods to explicit statistical, social-choice and debate principles more convincingly than v1. In particular, the group-selection quality bound, separate judge training, fixed-rate audits and explicit failure regimes make the research plan easier to assess. The remaining weakness is that RG3 still converts conditional statistical identities and tool-based screening into broader claims of truth-aligned learning, while the audit attribution and harm-rate claims outrun the specified model.

Naval relevance: Good. The proposal names the appropriate ONR S&T Focus Area and now evaluates missed and false departure calls at each audit budget, which is a useful improvement. The copied-report/compromised-agent example remains a credible analytical failure mode, but a replayed port record does not by itself justify a present-time vessel assessment or a patrol decision. The proposal supports broadly applicable basic research; its naval argument needs a bounded assessment task rather than stronger operational claims.

Mathematician/statistician: Good, borderline. RG1 now has a coherent mathematical rationale; the half-quality result is sound under its stated ranking/positive-weight assumptions. The largest concern is the asserted link between message credit and truth, followed by unqualified attribution and harm claims in auditing.

AI researcher: Good, borderline. Judge training, tool access for the single-judge baseline, frozen references and topic monitoring address real weaknesses. The remaining empirical question is which improvement comes from the new collective mechanism rather than stronger verification, and whether the proposed reward actually changes the incentives claimed for it.

Navy S&T reviewer: Good, borderline. The focus-area alignment and source-dependence problem are relevant. Decision consequences, verification cost and the source of truth in the scientific testbed need clearer limits, and the present document exceeds the technical page limit.

Decision on scientific content alone: borderline, improved from my v1 decline. This is not a prediction of an award: the NOFO also equally weights CV, support letter and budget. **One scientific change most likely to help:** remove the unsupported positive-multiple claim for message credit and make evidence-gated credit an explicitly tested method. **First submission action:** get the complete narrative, including required Sections 3.6-6, within 20 compliant pages.

C2. Keep these improvements; close the resolved v1 objections

Location: pp. 7, 9-13, 15-20, 22; eqs. (1), (4)-(6).

Change	My v2 judgment
Fixed human audit rate rho, rather than a fixed total budget	Resolves the previous budget inconsistency. Expected audit work is rho times the number of decisions; do not reintroduce the old fixed-total claim.
Jaffe attribution and development-set truth orientation, pp. 9-10	Substantially resolves the earlier model-credit and latent-label objections. The new statistical targets are now explicitly left open.
Half-split representative selection, positive weights and rating-independent fallback, pp. 11-12	Improves the old 1/K guarantee to 1/2 under the stated ideal ranking assumptions. The previously empty candidate fallback now has an action.
Judge training and qualification, pp. 13-15	A substantive answer to “why trust the judge?” Keep it. The empirical references support the choice to train local rulings, although competence under adaptive attacks remains a research question.
Tool-using single-judge baseline, p. 20	Addresses the most consequential missing baseline in v1.
Frozen references, weight applied after GRPO normalization, topic checks, pp. 18-19	Meaningful implementation clarifications. They do not prove the remaining RG3 claims, but should not be edited away.
Actual prior releases in Gold Standard Science, p. 22	Resolves the previous generic assertion of past practice.

Preserve three strengths: the one-question/three-RG architecture; the task-by-task approach/evidence/evaluation organization; and the adversarial evaluation with correct minorities, damaged majorities and explicit failure conditions. There is no reason to reorganize the whole proposal or restore the removed Propositions 5-6 as numbered theorems.

2. Eight problems worth addressing

C3. Rank 1 - four excess pages are a real submission blocker

Location: contents page; Sections 3.6-6, pp. 21-24; figures and tables throughout. **Must-fix.**

Reviewer concern: Sections 1-6 occupy 24 pages, not 20. The latest contents page now acknowledges the limit rather than claiming compliance, but the PDF remains four pages over it. Ending the upload at page 20 would omit the schedule/goals, Gold Standard Science, management, qualifications and responsibility material. Those sections are required narrative content, not excluded appendices.

Fix: complete a compression pass and then check the actual PDF, not the source's 12-point document-class setting. The current source still has 11-point nominal captions/tables and roughly 9-10-point diagram text; diagrams use Helvetica and body text embeds a Times-compatible face. The supplied NOFO says “Times New Roman, 12-point” and states no figure/table exception. Written grants-office or ONR clarification could settle an acceptable exception; older examples do not establish one. Ordinary 12-point TeX text extracting near 11.96 PDF points is unit conversion, not an independent violation.

Cuts that fund this change: remove the small conceptual overview Figure 1 and retain the more informative running-example Figure 2; remove Table 1, which duplicates Section 2 and overstates categorical novelty; remove illustrative Figure 7, whose bars are not results. Then shorten repeated gap/challenge/overview prose, repeated explanations following familiar equations, generic harness descriptions, and duplicated PI service/achievement prose. C34 gives a concrete cut ledger. Retain the quantitative simulation Figure 3 if its results are substantiated, the role mechanism Figure 5, the judge-training paragraph and the assumptions.

Page budget: recovering four pages requires substantial deletion, not a few sentence edits; larger compliant figure text may require further cuts. Target about 1,700-2,200 words of repetitive prose plus the redundant graphic areas, then recompile. This is an estimate, not a promise of an exact page yield. Do not shrink margins, move required narrative into the CV, or spend this space on additional method exposition.

C4. Rank 2 - three pilot passages are still explicitly unmeasured

Location: Tasks 2.1, 2.2 and 3, pp. 15, 17, 19; `pending` macros in the source. **Must-fix before submission.**

Reviewer concern: the numerical replay, audit and RL pilots are still orange `pending` material with exact past-tense results. The final-build flag is false. The two RG1 pilots are now ordinary black text; I cannot infer from the color change whether they were actually run. The README still calls five pilots open and acknowledges that the simulation script is absent, so the documentation is stale or the evidence status is unresolved.

Fix: retain only actual measured results supported by existing logs. For any unrun pilot, replace its entire passage with the prospective version in C22. If the first two have now been run, give the minimum actual setup in the existing passage: model identities, fit/development/test

separation, what constitutes a run, and what “0.92 quality” measures. I did not find scripts/logs in the inspected proposal folder; this is a verification limit, not evidence that the results are false.

Cut/payment: the prospective passages are shorter than the three orange passages they replace. For completed RG1 pilots, pay for setup information by cutting the repeated description of the same simulation in the RG1 Challenges paragraph, and the narrative interpretation “so the hint did not decide for the group.” Do not add more precision or seed counts than were actually measured.

Final action: replace/remove every pending result and enable `finalbuildtrue`. The guard only catches remaining macros; it cannot certify unmarked numbers. Do not turn an unrun result black to make it pass the guard.

C5. Rank 3 - the new message-credit truth-alignment assertion is false

Location: Task 3, pp. 18-19, “Which debates count,” eq. (6); overview p. 3. **Must-fix.**

Reviewer concern: restricting credit to tool-backed disputes is a sensible screening rule. It does not imply that, whenever accepted decisions are right more often than wrong, expected message credit is a positive multiple of movement toward truth. Final-decision errors can depend on precisely the message whose credit is being measured. Neither replay nor marginal accuracy supplies the needed independence. The latest comparison to RLVR does not repair this: verifying one disputed step does not verify the final label or every rewarded confidence increment.

Concrete counterexample: suppose 80% of accepted decisions are correct. In those debates, a message increases the judge's probability of the true/final answer by 0.01. In the remaining 20%, a misleading message increases the probability of the wrong final answer by 0.40. Average eq. (6) credit before its common weight is 0.8*0.01+0.2*0.40=+0.088, while average movement toward truth is 0.8*0.01-0.2*0.40=-0.072. All final confidences can be 0.80, consistent with aggregate calibration and the same positive weight. Replay can succeed in every case: a missing premise or stale source can still make the decision wrong. The counterexample meets the proposed screening and better-than-chance conditions.

Replace the sentence beginning “But it rests on evidence rather than persuasion” with:

> Restricting credit to tool-backed debates removes one failure mode but does not establish truth alignment: an authentic output can support a misleading inference. We test whether this restriction reduces reward hacking relative to debate-win and final-decision rewards.

Cut/payment: replace that roughly 50-word claim with this roughly 40-word statement; merge the following repeated comparison sentence into it. Keep the preceding caveat that replay settles only the disputed step. Retain eq. (6) as a candidate process reward, not as a proved truthful reward. No new experiment or theorem is needed to make the text correct.

C6. Rank 4 - the chance-correction baseline does not deliver the claimed training incentive

Location: Task 3, pp. 17-18, eq. (5), “Why this method”; Section 1.2, p. 3. **Must-fix the claim, not necessarily the objective.**

Reviewer concern: the draft now correctly calls the cross-question subtraction a baseline and fixes references during update phases. But an action-independent detached baseline does not change the expected scored-answer policy gradient. If it is shared across GRPO completions for a question, it cancels exactly in the centered advantages. Thus “uninformative policies have expected score zero” is not itself a proof that this baseline prevents training from converging to conformity.

For completions i with match indicators I_i and a common baseline b_q, r_i=omega_q*I_i-b_q gives normalized advantages (I_i-mean(I))/sd(I), for positive omega_q and nonzero variance. Applying omega_q afterward, as the revision proposes, preserves that weighting; it does not restore a distinct effect of b_q. Independently resampling a detached baseline can change variance but still supplies no extra expected scored-action gradient in the elementary policy-gradient argument.

Replace the baseline/zero-score explanation with:

> The cross-question term centers the reported agreement score; when detached, it is an action-independent baseline rather than an additional incentive guarantee. The learning rationale rests on the frozen leave-group-out reference under the conditional noise assumptions. We test whether those assumptions remain useful during training.

Cut/payment: replace the existing two sentences beginning “The second term” and “A wrong decision thus shrinks,” shortening the later explanation of self-influence to retain the omega/GRPO implementation detail. On p. 3 replace the sentence beginning “As in learning from noisy labels” with “A conditional noisy-label identity motivates first-answer training; held-out tests and random audits check when its assumptions fail.” This substitution is shorter.

Keep the distinction: the per-question binary identity is valid under its assumptions. Marginal topic error below one half neither establishes independence nor certifies all questions. A topic-level stop rule can detect some failures; it is not a proof that none occur before detection. The proposal already admits delayed reinforcement of shared mistakes on p. 21; the overview should not promise otherwise.

C7. Rank 5 - audit attribution and harm rates are asserted before their model is specified

Location: Task 2.2, pp. 16-17, “Assigning blame and flagging” and “Bounding the harm”; Table 4 G4, p. 22. **Must-fix the assertion; keep the research target.**

Reviewer concern: random pairings do not alone identify individual malicious contributions when outcomes include partner interactions, selected task difficulty and joint failures. An anytime-valid test controls false alarms only under a valid conditional null; it does not create that null, identify blame, or distinguish dishonesty from honest weakness. The proposed square-root harm rate is a plausible target under an appropriate model, but the cited confidence-sequence paper does not establish this system's attribution or harm guarantee.

Replace the attribution opening with:

> We will model excess error across randomized role assignments and state conditions under which individual and coalition effects are identifiable. Flags indicate excess error relative to a specified honest baseline; they do not by themselves establish intent.

Replace the strongest harm claim with:

> Under that model, we will study whether excess wrong decisions before flagging admit a bound of order sqrt(n log log n/rho), with dependence on the honest error level, false-flag probability and per-agent exposure. We will separately analyze the zero-honest-error case, where detection can require a bounded number of attacks at fixed rho.

Cut/payment: these passages replace the existing attribution assertion and the current bound explanation/tax-audit analogy; the combined replacement is shorter. The small expression is a conjectured target, not a theorem inherited from Howard et al. (2021). Replace “the slack any anytime-valid test must allow” with “a candidate scaling suggested by time-uniform concentration.” Replace “when they rarely do” with “when honest errors are zero and audits are correct.” A small positive honest error rate is not equivalent to zero and need not permit a constant harm bound.

This is worth retaining as basic research: characterize the tradeoff and its failure regime. It is not necessary to prove it before submission. The fixed-rate resource model itself is now coherent.

C8. Rank 6 - the evaluation still conflates verifiable claims with review preferences

Location: Section 3.5, pp. 19-21; Table 2, p. 20; Table 4 G1/G3/G5, p. 22. **Should-fix; ground-truth wording is essential.**

Reviewer concern: “Every question has a known answer” is not established for scientific support claims selected from OpenReview or PeerRead. A reproducibility report resolves particular executable claims, not all assertions in the associated paper. Unsettled decisions are excluded from training, but their treatment in accuracy is unspecified; selective abstention can make the system look better without doing more useful work.

Replace the Section 3.5 opening with:

> Math/code and synthetic maritime questions have reference answers. In testbed C, truth-based accuracy is restricted to independently verified executable claims or controlled planted errors; agreement with review scores is reported separately. We report decision error and abstention coverage, including unresolved disputes.

Cut/payment: replace the existing opening, shorten Table 2 C's corpus-description cell, and delete the redundant caption claim “Each provides known answers.” This is about the same total length. Replace “RLVR as an upper bound” on p. 19 with “matched-compute training on reference answers as a supervised reference”: it is a useful benchmark, not a mathematical upper bound on achieved accuracy.

Comparators: keep the newly added tool-using single judge; that fixes a major v1 weakness. If only one additional planned comparison is affordable in the narrative, name the dependent-classifier model directly in Task 1.1, because your generative hierarchy adapts it. A

comparison only to the independent model cannot establish which improvement is new. Peer-prediction and cohort training remain relevant future comparisons, but I do not ask for any new predeadline runs. Replace “all baselines” in G3 with “the matched-budget tool-using judge and open discussion.” In place of generic cost prose, name Brier score for probability quality and a fixed-bin calibration error with its binning rule; Brier score alone does not isolate calibration.

C9. Rank 7 - the Navy story's assessment time and decision consequence remain too strong

Location: Section 1.1, p. 1; Figure 2, p. 2; Section 1.4, pp. 4-5; naval metric p. 21. **Should-fix.**

Reviewer concern: the copied-source/compromised-agent failure is credible. But the example's 09:10 port record does not establish “still in port” at an unspecified later time, and the narrative makes a majority vote itself send a patrol. That invites an avoidable objection from a Navy reader, without strengthening the basic research.

Exact local substitutions: change the question to “Was V out of port at 09:10?” and the consequence to “A majority vote produces a false departure assessment that could misdirect patrol tasking.” Change Figure 2's final decision to “Assessment at 09:10: the port record places V at berth 4; the officer retains tasking authority.” State that the scenario is illustrative, not a validated Navy workflow.

Cut/payment: these substitutions replace longer existing sentences. In the naval section replace the broad maintenance/logistics/cyber/autonomy bullet with a shorter statement connecting copied/stale/forged reports to false and missed assessment rates under a fixed verification budget. The new metric on p. 21 is the right direction; move its purpose into the existing Navy paragraph rather than adding another use case.

The title can stay. Preserve broad applicability through source dependence, uncertain truth and verification budgets. Do not turn the project into a ship-tracking demonstrator or add an unverified transition partner.

C10. Rank 8 - the support boundary remains too generic for the newly expanded judge/reward work

Location: Section 4, p. 23; judge training p. 15; RG3 pp. 17-19; scientific testbed pp. 19-20. **Should-fix; verify the factual boundary.**

Reviewer concern: the current paragraph still says this award supports aims the other awards do not, without saying which. The pending hypothesis proposal develops execution-based validation, reward models and agent training; the complex-reasoning proposal develops explanation/QA evaluation. The expanded judge-training plan makes shared infrastructure easier to mistake for another proposed scientific aim. Open Philanthropy support for debate-control evaluations is potentially closer still, but its scope is not supplied.

Replace the NSF portion with:

> CAREER supports event extraction and graph construction; ONR reuses the resulting question-generation infrastructure. The pending NSF reasoning and hypothesis proposals study explanation methods and scientific discovery/implementation. ONR studies collective decisions under dependent errors, impartial role selection, adversarial audits and multi-agent learning from noisy references. Shared infrastructure will be distinguished from ONR-funded aims and effort.

Cut/payment: replace the current support paragraph's generic assurances and cut one repeated award/service sentence from Section 5 if needed. Retain a separate short factual sentence distinguishing the actual Open Philanthropy and Amazon work after checking their scopes; I cannot invent it. No implication of duplicate funding is made from topic similarity alone. C25 records the reference-file comparison.

3. Mathematical audit: what I verified

C11. Equation (1): the calibration residual repairs the earlier logical gap

Location: Section 3.1, pp. 7-8, eq. (1). **Keep; low-cost clarification only.**

Let A={pi>=tau} and e(p)=P(leading answer wrong|pi=p). The no-debate contribution is E[e(pi)1_A]. Splitting e(pi)=(1-pi)+(e(pi)-1+pi) and bounding the latter by its positive part gives exactly the displayed no-debate upper bound. Adding the routed debate union bound yields (1). This no longer requires perfect calibration.

Define the epsilons on the routed population, with suitable conditioning on preceding failures being absent, or call them upper bounds for those conditional probabilities. That is a clarification, not a new method. Replace the sentence “Removing a task leaves its term uncontrolled” with this conditioning statement; its causal necessity claim is stronger than the decomposition establishes anyway.

Assumptions/design/novelty: a standard probability decomposition organizes where routing, candidate coverage, role integrity and judging enter. It is load-bearing as error accounting, not new mathematics or a computable adversarial guarantee. Estimating the positive-part conditional calibration residual from sparse audits requires an estimator/binning or an upper bound; random audits alone do not specify one. This can remain part of the research. Do not add an estimator derivation at the deadline.

C12. Equation (2) and Proposition 1: correct under the shared-factor model

Location: Task 1.1, p. 9, eq. (2), Proposition 1. **Keep.**

The model is truth -> one latent answer per group -> conditionally independent member outputs. Summing over the latent answer gives G_g; multiplying independent groups gives the likelihood in (2). For binary labels and lambda in (0,1/2], the group likelihood ratio is `R(Lambda)=((1-lambda)*Lambda+lambda)/((1-lambda)+lambda*Lambda)`. Its derivative has the sign of 1-2*lambda; its endpoints are lambda/(1-lambda) and the reciprocal. Therefore the absolute log-odds contribution is at most log((1-lambda)/lambda), regardless of group size. The symmetric multiclass pairwise bound is also valid under the corresponding truth-dominant channel.

This is a useful cap on redundant evidence. It assumes dependence is mediated by the specified factor; it does not establish that real source overlap, changing topics or collusion follow that structure. The current text now says this and credits Jaffe et al. (2016), which materially improves the presentation. The exact saturation argument is an elementary channel calculation; I did not verify priority for that formulation.

Minor, not a funding problem: write “absolute log-odds” and 0<lambda<=1/2; the zero-error endpoint otherwise gives an infinite bound. The effective group distribution may be identifiable without separately identifying lambda and member noise. Your main decisions use the effective likelihood, so do not spend space on that parameterization caveat unless interpreting a learned lambda as source reliability.

C13. Proposition 2: correct three-view principle; the observation wording is slightly too broad

Location: Task 1.1, pp. 9-10, Proposition 2. **Low-priority tightening, not a structural objection.**

With K truth classes, positive class weights and three jointly observed conditionally independent full-rank views, Kruskal's condition is 3K>=2K+2 for K>=2. The effective group response distributions are identifiable up to a common label permutation. The draft now explicitly uses separate known-answer development data to orient/calibrate them; that closes the earlier headline ambiguity.

The current phrase “each group ... jointly with two other groups” permits disconnected triples. With balanced truth, six groups observed only as two disjoint triples can have the second triple's effective accuracies flipped from 0.8 to 0.2 without changing any observed distribution. Each triple can independently reverse its latent labels. This is not a counterexample to the three-view theorem; it is a missing observation-connectivity condition.

Small replacement if convenient: replace the observation clause with “the groups are jointly observed in overlapping full-rank triples that align a common truth labeling.” Interpret it with a stationary response model and ignorable randomized observation. Do not add the counterexample to the proposal. The binary explanatory example is correct for nonzero factors: u_ij=2P(agreement)-1=(2p_i-1)(2p_j-1), so (2p_i-1)^2=u_ij*u_ik/u_jk, with a shared sign ambiguity.

Credit/design: properly attributed to Kruskal (1977)90069-6) and Allman et al. (2009). It motivates independent views; it does not prove learned partition recovery or robustness. The “what remains open” paragraph handles that distinction well. Remove H1's claim that empirical advantage necessarily disappears below three groups: replace it with “the identifiability guarantee no longer applies,” at a word saving.

C14. Equation (3): the decision-theoretic derivation is correct

Location: Task 1.1, p. 10, eq. (3). **Keep; qualify “exactly.”**

For calibrated leading-answer risk and a specified debate error epsilon_D, accepting costs L(1-pi) and debating costs c+L*epsilon_D. Comparing them yields pi<1-epsilon_D-c/L. Raising L raises the threshold when the other quantities are held fixed. This is an expected-loss/reject-option principle, now appropriately credited to Chow (1970); it is not a new general optimal-routing theorem.

An average development error need not equal the conditional debate error on a specific type of question. If a certain regime has pi=0.80, c/L=0.01 and true debate error 0.40, a global estimate epsilon_D=0.05 would route it to debate although its true expected cost is worse. Replace “worth its cost exactly when” with “under this constant-error approximation, worth its cost when.” The already planned stratification by tool-settleability is useful. A broader conditional risk model can be future work rather than extra deadline text.

The claim about unanimous cases should say that the posterior corrects dependence-induced overconfidence. Low-confidence routing using that same corrected posterior could route them too. Replace the comparative sentence, rather than adding a new distinction; this is a wording correction, not a reason to redesign Task 1.1.

C15. Equation (4) and Proposition 3: the new half-quality argument is sound

Location: Task 1.2, pp. 11-12, eq. (4), Proposition 3. **Keep.**

Part (a): conditional on fixed initial answers, groups, eligibility and historical weights, a group's current ratings only affect choices in the other half. They cannot change its own members' representative or judge probabilities. The fallback is now independent of ratings, including when it assigns an agent an answer different from its original one. This resolves the previously undefined empty-candidate fallback.

Part (b): condition on the drawn judge and an answer supported by at least two nonjudge groups. The redraw condition is invariant to swapping halves. Thus the best eligible agent is in its answer's candidate half with probability exactly 1/2. On that event, same-answer raters exist across the split; every positive-weight rater ranks by the same quality, so the weighted rule selects a highest-quality candidate. Nonnegative quality supplies the lower bound on the complementary event. I also checked swap symmetry by enumerating 360 two-answer support patterns for two through five groups. This checks the finite combinatorics, not empirical quality.

Part (c): for any lottery on n_q eligible judges, the f largest probabilities sum to at least f/n_q. Uniform sampling attains that worst-set bound. Correct: it is minimax occupancy protection for a fixed eligible pool, not a reliability theorem. The step-trained, qualified pool is the separate empirical reliability measure.

Credit/design: the partition lineage is credited to Alon et al. (2011), with impartial selection context from Kurokawa et al. (2015). The adapted half-quality argument is more directly useful than v1's 1/K group lottery bound; no exhaustive originality claim is established here.

Do not reopen as major objections: with fewer than two nonjudge groups the split needs an availability guard; fallback agents can sometimes both rate and be assigned a role, so “either rates or may be chosen, never both” is too absolute. The invariance proof survives. Replace that slogan with the proposition's actual current-rating invariance, at the same length; handle insufficient groups as an unsettled input. These are small implementation qualifications.

C16. Proposition 4 and the new k*epsilon_J bound: valid idealization, useful only with matching error conditioning

Location: Task 2.1, pp. 13-15, Proposition 4 and following bound. **Keep.**

If the correct advocate supplies true steps, actually disputes a false step of the wrong argument, and the judge evaluates it correctly, the losing rule selects the correct answer. This is the inherited debate argument, credited to Irving et al. (2018) and Brown-Cohen et al. (2024). It does not prove that a bounded exchange finds that step.

For at most k rulings, a union bound gives failure <= failure of the argument conditions plus k*epsilon_J, provided epsilon_J bounds each encountered ruling's error under the relevant nonsteering/valid-candidate conditions. No independence between rulings is needed. With adaptive recursion, use a fixed ruling cap or an appropriate conditional bound; an average unchallenged development error is not automatically such a bound. These are assumptions to state compactly, not a new proof to add.

The trained-judge paragraph is the real improvement. Ling et al. (2023), Table 3, supports the reported 52% versus 69% verification comparison in its structured format/setup. Tyen et al. (2024) separates error finding from correction. Neither establishes that the opposing side always finds the error, but they justify the experimental design. Keep the cited observations rather than claiming universal local-judging superiority.

Local issue worth one replacement: “The first disputed step is enough” is not guaranteed for general arguments. An adversary can dispute a true early step; upholding it does not certify later steps. Replace that sentence with “Each challenger searches for a decisive false step; failure to find one within budget contributes to condition (ii)'s failure probability.” This is shorter and aligns the protocol with the displayed bound.

C17. First-answer noisy-label identity: correct conditional fact; not a theorem about all learning updates

Location: Task 3, p. 18, eq. (5), “Why this method.” **Keep the identity; apply C6.**

For fixed q, truth y and scored action x, if the frozen binary reference has error eta_q independent of the action, P(D=x)=eta_q+(1-2*eta_q)1{x=y}. Multiplying by the scored action's policy score and taking expectations removes the additive constant, giving the stated per-question REINFORCE scaling. An action-independent weight can multiply that identity. This is elementary random-label-noise reasoning, consistent with the lineage of Natarajan et al. (2013); that paper is not a proof of this multi-agent GRPO system.

Group exclusion removes direct self-inclusion. It does not guarantee statistical independence after missed dependencies, shared training or strategic cross-group coordination. Marginal label error below 1/2 is not enough. For example, if X=D and both are wrong on the same 40% of questions, copying achieves agreement 1 while truth achieves 0.6. This violates independence, precisely the condition needing measurement.

The full shared-parameter update also need not point in the same direction as the truth-trained average: equally likely questions with true-reward gradients +1 and -2 and errors 0.01 and 0.49 give true mean -0.5 but noisy mean (0.98-0.04)/2=+0.47 before omega. Applying omega=1-2*eta makes the noisy mean +0.4798, still opposite. These are locally realizable correct-answer probabilities 0.5+theta and 0.5-2*theta. This does not invalidate the conditional identity; it shows why that narrower p. 18 qualification should control p. 3's overview.

Freezing references and applying omega after GRPO normalization are useful practical choices. Do not infer that the identity predicts final accuracy gains, all GRPO/KL/clipping behavior, or the full chance-corrected objective's gradient. Topic audits and held-out tests monitor violations; they do not eliminate them by assumption.

C18. Equation (6): telescoping is correct; truthful credit is not established

Location: Task 3, pp. 18-19, eq. (6). **Keep as a tested reward after C5.**

For a fixed final label D and fixed omega, summing eq. (6) over messages telescopes to omega*[P_J(D|full transcript)-P_J(D|empty transcript)]. This assigns order-dependent confidence increments. It is neither a leave-message-out causal effect nor automatically a truthful contribution score. The draft correctly says policy invariance from Ng et al. (1999) does not carry over; retain that qualification.

The counterexample in C5 disproves the remaining positive-multiple claim even with accepted-label accuracy 0.8 and calibrated final confidence 0.8. Correctness of D must be sufficiently independent of the confidence increment for the sign-flip identity to transfer, and D is produced from the same debate. A repeated call authenticates an observation; it does not provide that independence. This is the principal remaining mathematical error, not an obscure parameter edge case.

Design justified: test whether replay-gated process credit outperforms terminal or win rewards. **Scientific impact if successful:** an empirically supported and characterized credit mechanism. It need not be an unconditional truth-alignment theorem to be worthwhile basic research. Delete the unsupported implication rather than abandoning RG3.

C19. Audits: exposure formula correct; attribution and harm require additional theory

Location: Task 2.2, pp. 16-17. **Apply C7.**

For a specified n attack attempts committed before independent post-decision Bernoulli audits, zero exposure has probability (1-rho)^n. Predicting planted cases does not predict the next independent real-audit draw. This exposure fact is now ordinary prose with a citation; there is no reason to restore it as Proposition 5.

With m approximately rho*n audited decisions and bounded honest residuals, a time-uniform statistical boundary of roughly sqrt(m log log m) suggests a decision-scale error of roughly sqrt(n log log n/rho) after dividing by rho. This makes your target plausible under a suitable exposure/null model. It is not yet a bound on an individual colluder in a role-selected system with estimated partner effects, noisy truth and coalitions. Its constants and coverage must depend on alpha and the exposure/error assumptions.

“Rare honest errors” do not imply a bounded number of hidden attacks: any fixed positive honest error probability leaves noise that can mask small continuing excess error. In the ideal zero-error/perfect-audit case, the first malicious error can be detected on audit; expected attacks to exposure are 1/rho, with tail dependence on the desired failure probability. That is the precise favorable boundary to analyze.

Howard et al. (2021) provides confidence-sequence tools, not a direct theorem establishing all these system-specific conclusions. The most defensible basic-research target is to derive the missing conditions and characterize when the proposed rate fails.

4. The ad hoc test and the original questions

C20. Per-task judgment and colleague-question status

Location: Sections 3.2-3.4, pp. 8-19.

Task	V2 assessment	Remaining decision-relevant gap
1.1	Principled generative model and expected-loss routing; appropriate predecessor attribution; learned partitions explicitly left open.	Establish finite-sample calibration/routing performance under specified deviations. This is a legitimate research target, not a missing deadline experiment.
1.2	Principled current-rating impartiality with a meaningful conditional 1/2 quality guarantee.	Honest common quality rankings are an idealization; coalition behavior and realized debate quality remain empirical/theoretical targets.
2.1	Local-dispute principle now connected to a concrete judge-training plan and a failure decomposition.	A trained local verifier is not guaranteed to find or correctly judge an adaptively selected dispute; measure those failures.
2.2	Principled independent audit sampling; the attribution/test model remains underdefined.	C7: formulate the honest null, exposure conditions and coalition effects before promising individual harm bounds.
3	Principled frozen-reference binary identity; process credit remains an empirical candidate.	C5-C6: tool-backed selection does not establish message-credit truth alignment; detached chance subtraction is not an extra gradient incentive.

The PO's “ad hoc” objection: substantially better answered for RG1 and the RG2 judge design. It remains partly unanswered for attribution and message credit. The remedy is a sharply posed research question and accurate conditional claims, not more decorative propositions.

Colleague (a), reliability without ground truth: partly answered, pp. 9-10. Fixed-group identifiability and truth orientation are explicit; arbitrary collusion/learned group recovery are not solved and are now identified as open.

Colleague (b), why trust the judge: substantially answered as a research plan, pp. 11-15. Independent training targets, qualification, a pool and subsequent audits provide a concrete rationale. They support testing trustworthiness, not an unconditional guarantee against evaluation-aware colluders.

Colleague (c), wrong group decisions reinforcing behavior: partly answered, pp. 18-19 and limitation p. 21. Frozen references and monitoring are improvements; average accuracy alone cannot establish the needed independence, and message credit still overclaims.

Colleague (d), concrete metrics: mostly answered at the metric-family level, pp. 19-22. C8 adds the two material missing definitions: what has a truth label, and what happens to unsettled cases.

The PO's six questions are visibly answered: problem on p. 1; importance on pp. 1-5; current methods on pp. 5-6; approach/newness on pp. 3-19; limits on p. 21 and within tasks; impact on pp. 4-5 and 21. The issue is no longer missing sections. It is whether the strongest overview claims follow from the detailed qualifications.

C21. Compared with the approved white paper: commitments preserved, learning reference changed

Location: repository four-page v4 white paper pp. 2-4; full proposal pp. 7-19.

The central question, three RGs/five tasks, weighting, impartial roles, evidence-replaying debate, planted/real audits, three testbeds and red team are preserved. The full proposal adds formal models, a different half-split mechanism, judge training, calibrated-error accounting and frozen learning references. No approved task appears dropped.

The main substantive change remains RG3: the white paper trains on final group decisions; first answers now train against a leave-group-out posterior answer, while only tool-backed debates train message credit. That distinction is scientifically defensible but the caption/overview still says “RG3 trains every agent on the decision.” Replace that phrase with “RG3 trains first answers against other groups and messages on tool-backed debate outcomes.” Pay its few extra words by deleting the redundant feedback-arrow label. It avoids giving the PO an inaccurate account of the revised method.

I would not restore omitted numerical examples or the white paper's strongest scaling language merely because the PO approved its structure. Approval of a white paper is not validation of every added mathematical claim.

5. Pilots: evidence status and deadline choices

C22. Keep measured evidence; make unrun pilots plainly prospective

Location: preliminary results pp. 10-12, 15, 17, 19; Figure 3 p. 8.

Pilot	Current source status	Would the claimed result convince this panel?
1. Shared-factor aggregation	Ordinary text, pp. 10-11; formerly pending.	Supports feasibility under seeded dependence if measured with clean fit/development/test separation. Does not establish natural dependence recovery or improvement over the inherited model. No independent logs supplied.
2. Impartial selection	Ordinary text, pp. 12-13; formerly pending.	Zero gain checks implementation of rating invariance; the 0.92 quality ratio is useful only with a defined nonnegative quality scale. Elo itself has an arbitrary origin, so an unqualified Elo ratio is unsuitable.
3. Forged-output replay	Still pending, p. 15.	Strong evidence for execution authentication if run. It isolates replay rather than the full benefit of debate; the tool-using judge is now appropriately in the future baseline plan.
4. Plant detection/audit flagging	Still pending, p. 17.	AUC and median audit counts would show behavior of the tested discriminator and simulation, not undetectability or universal false-flag validity. Need audit rate, null error, repetitions and per-agent exposure from the existing pilot.
5. RL stability	Still pending, p. 19.	A measured learning curve would support eq. (5) under that training setup; it does not test eq. (6), and one curve cannot establish general noncollapse. The 73% gain comparison is descriptive if measured, not a theoretical prediction.

If pilots 1-2 are complete, do not rerun them just to satisfy this review. Verify the records and replace the stale README status. If they are unrun, present them prospectively as well. Figure 3's generating script is still absent from the inspected folder/acknowledged in the README; retain numerical simulation claims only if its results can be substantiated from existing work.

For an unrun pilot 3, replace its entire orange passage with: “Planned pilot: on 200 tool-settleable disputes with forged outputs, compare transcript-only judging with exact-call replay, measuring forgery acceptance, decision accuracy and verification cost.”

For an unrun pilot 4: “Planned pilot: compare detectability of templated and controlled-edit plants, then simulate excess-error flags at specified audit rates, reporting total decisions and audits to flag, harm before flagging and false flags.”

For an unrun pilot 5: “Planned pilot: train a 1.5B model with flawed peer references at matched compute, comparing eq. (5), majority-agreement and reference-answer rewards using held-out learning curves and collapse indicators.”

Each replacement is substantially shorter than its original passage and funds compression. Do not add them alongside the numbers. If pilots 1-2 are incomplete, their prospective statements should likewise name setup and measured outcomes without guessing results.

Priority with the remaining day: validate existing evidence first. Among the remaining pilots, the replay pilot is the most contained feasibility test; use an RL result only if reliable training logs already exist. The unresolved message-credit guarantee should be corrected in prose rather than “validated” by an unrelated first-answer pilot. Do not launch a sixth experiment or delay a compliant submission to complete every pilot.

6. Navy relevance, skim test and funded-example calibration

C23. What a busy panelist will carry into discussion

Location: title, p. 1, Figure 1; running example Figure 2 p. 2; Navy section pp. 4-5.

The title matches the intended research program and can remain. The new small overview makes the RG cycle visible; the detailed running example is still the more informative figure and should survive compression. If one figure must go, delete the redundant overview rather than the method/evidence figures.

Three-sentence skim impression: “The project asks how a group of AI tools can reach an auditable answer when they copy mistakes and some are compromised. It combines dependence-aware weighting, impartial debate roles, evidence verification, random audits and learning from peer decisions. The mathematical rationale is clearer than before, but trustworthy learning still depends on assumptions the proposed safeguards do not yet establish.”

The first non-specialist stumbling point is still why two debaters plus verification are needed when a single database query resolves the example. The new tool-using single-judge baseline is the correct scientific answer; make that comparison explicit in the existing motivation sentence. Replace “We propose debate as a foundation” with “We test when debate exposes decisive checks more effectively than a tool-using single judge.” This shorter substitution retains ambition while stating the actual research question.

The Navy link is credible as basic research on source-dependent assessments. The five broad use-case bullets are less valuable than the revised per-100-case missed/false departure metric. There is no need to invent a Navy collaborator, deployment commitment or further operational detail.

C24. What the funded examples still do more convincingly

Location: your pp. 8-22; funded examples' technical sections only, locations below.

Calibration	Panel-visible difference	Worth changing now?
Joshi PN-6 to PN-8, eqs. (1)-(4), Figure 2	A quantified improvement in a mathematical object precedes explicit new research directions. Your model assumptions are clearer, but the promised new guarantee is diffuse across several tasks.	Yes: keep the existing “what remains open” statements, and describe calibration/routing error and audit harm as the new targets. Do not add another theorem or proof at the deadline.
Joshi PN-3 to PN-4; Yousefi technical p. 4	The naval constraint changes the optimized quantity, rather than only supplying an application list.	Yes: use C9's verification-budget assessment framing and the metric already added on p. 21. Replace broad applications; no new scenario.
Chen technical pp. 12-13 and 19, evaluation plans	Conditions and measures specify what successful assistance means.	Yes: define truth versus preference and abstention coverage through C8. Your schedule is already specific enough.
Yousefi technical pp. 18-19, Table 1/Figure 8; Zhang technical pp. 16-18	Completed evidence has identifiable experimental conditions and concrete outcomes.	Yes: resolve pilot status and preserve actual setup details. Do not imitate visual density or equation count.

I do not recommend changing the section order, adding a large human study, expanding naval demonstrations, or copying wording from these examples. The page-limit problem makes selective deletion more valuable than further exposition.

7. Overlap and FY27 compliance

C25. Overlap remains localized; the technical paragraph still needs a truthful boundary

Location: Section 4 p. 23; testbeds B/C pp. 19-20; reward and judge training pp. 15, 18-19. NSF reference locations below use physical PDF pages.

Funded CAREER, reference PDF pp. 2-3 and 6-13: event extraction, question generation, event graphs, consistency rewards and schemas. Reusing its outputs is sensible. Avoid presenting fresh development of that pipeline as a new ONR aim. The present reuse acknowledgment is helpful; C10 states the exclusion explicitly.

Pending complex reasoning, reference PDF pp. 6-9 and 11-15: event-based QA, explanation generation/evaluation, reward modeling and refinement/checker modules. Reusing those capabilities inside debate is distinct from funding their generic redevelopment. The current generic pending-proposal sentence does not state this boundary.

Pending hypothesis proposal, reference PDF pp. 9-16: scientific-review reward models, execution-based validation, ResearchGym, and multi-granularity agent training. Adjacency is substantial for a generic scientific sandbox or single-agent step reward. The distinctive ONR aims are collective inference, impartial roles, adversarial audits and multi-agent learning under dependent noise. C10 separates aims without claiming that generic infrastructure is novel.

These comparisons show potential overlap, not established duplicate funding. The Open Philanthropy debate-control scope and actual effort/cost allocations are not supplied and cannot be certified from these PDFs. Verify them using award records rather than inserting an invented disjointness assertion. Also verify the current resources line: it now says ten A100 GPUs, whereas v1 described six A100 and four RTX A6000; a real upgrade may explain this, but a rename would not.

C26. Compliance findings and exact review standard

Location: all technical pages, especially pp. 20-24; FY27 NOFO printed pp. 15-16, 19-20, 29-30.

Requirement	Current finding
Technical narrative <=20 pages	Fails in the reviewed PDF: Sections 1-6 end on numbered p. 24. Cover/contents/CV exclusions do not exempt management or responsibility.
Times New Roman, 12-point; 1-inch margins; letter size	Source sets appropriate geometry; captions/tables and diagrams remain smaller, with Helvetica diagrams and a Times-compatible body face. No stated small-font exception found in the supplied NOFO. C3 addresses this before optional edits.
Primary ONR S&T Focus Area and naval relevance	Present, pp. 4-5; the named area and associated research areas match the NOFO. No need to add a secondary focus area.
Schedule/milestones and goals/benchmarks	Present on pp. 21-22, currently outside the permissible 20-page narrative. Keep them inside the final compressed narrative.
Gold Standard Science	Present p. 22 with concrete prior releases, a clear improvement. Do not cut this required section to solve pagination.
Single-PI management, qualifications and responsibility	Present pp. 22-24; compress duplicated achievements rather than remove required topics.
Results and build status	Three `pending` passages remain; final-build flag false. Two former pending passages are now unmarked. Resolve actual evidence status.

The NOFO's technical-merit criterion reads:

> (1) Technical Merit - A creative research proposal, demonstrating the potential for making progress in an ONR research area and the degree of innovation, and soundness of technical concept. The Principal Investigator's awareness of the state of the art and understanding of the scope of the problem, significance and originality of the technical approach and effort needed to address/solve the problem and anticipated scientific impact within the field. The Principal Investigator's capabilities, related experience, facilities, techniques or unique combinations of these which are integral factors for achieving the proposal objectives; and the qualifications, capabilities and experience of key personnel and team members who are critical to achieving the proposal objectives. (Note: this is single PI award, no co-PI's allowed).

The scientific/alignment portion of criterion (2) calls for “new knowledge to develop new technologies that ultimately become new capabilities delivered by the acquisition community” and alignment to an ONR S&T Focus Area. The NOFO states that criteria 1-5 are equally important. This technical-only review does not evaluate the excluded CV, support letter, budget or disclosure forms.

8. Targeted line edits and compression

Use these as replacements, not additions. Where they duplicate a ranked fix, choose one version. I have omitted stylistic polishing that does not improve credibility, mathematical accuracy or page count.

C27. Shorten the operational claim

Location: Section 1.1 p. 1, example; Figure 2 p. 2.

Original: “A majority vote picks the wrong answer and sends the patrol to sea after a ship still at berth ...”

Revised: “A majority vote gives a false departure assessment that could misdirect patrol tasking ...”

Shorter; also time-index the question as C9 specifies. This is an illustrative assessment, not a verified Navy procedure.

C28. Make the overview match the conditional training result

Location: Section 1.2 p. 3, RG3.

Original: “As in learning from noisy labels ... occasional wrong decisions slow this training but do not reverse it, and audits stop it on any topic where they become frequent.”

Revised: “A conditional noisy-label identity motivates first-answer training; held-out tests and random audits check when its assumptions fail.”

Shorter. “No per-training-example truth labels” is accurate if you retain a label-use qualification; the judge training, orientation/calibration and audits do use trusted labels.

C29. Credit the existing model while retaining the new research claim

Location: Section 1.3 p. 3, novelty item (i).

Original: “a statistical reliability model for agents that answer and argue ...”

Revised: “dependence-aware debate routing built on a shared-factor reliability model ...”

Shorter. The body now correctly credits Jaffe et al.; the front-page novelty description should reflect that relationship without weakening the new routing/strategic research targets.

C30. Remove a confidence-comparison overclaim

Location: Task 1.1 p. 10, after eq. (3).

Original: “... a confident, unanimous answer ... can still fall below tau ... which rules triggered by low confidence or disagreement cannot do.”

Revised: “Dependence correction can lower the posterior of an otherwise confident unanimous answer enough to trigger debate.”

Substantially shorter; disagreement-only routing misses unanimity, but confidence routing using your corrected posterior need not.

C31. Align the mechanism slogan with the proposition

Location: Task 1.2 p. 12, “Why this method.”

Original: “An agent either rates or may be chosen, never both.”

Revised: “A group's current ratings cannot change its members' selection probabilities.”

Comparable length. The revised fallback can assign a former rater an opposing role; this does not invalidate the actual invariance result.

C32. Replace an unproved source of independence

Location: Task 2.1 p. 15, judge-training paragraph.

Original: “... judges are built on models other than those that propose false steps, so that a true step wrongly flagged is not also wrongly struck.”

Revised: “... judges use different models from the false-step proposers; we measure their joint errors rather than assume independence.”

Shorter. Model diversity is a safeguard to test, not a guarantee that two agents will never share a mistake.

C33. Name the scientific reference comparison accurately

Location: Task 3 p. 19, “Checking the premise.”

Original: “... train the same agents on the known answers (standard RLVR) as an upper bound ...”

Revised: “... compare with the same agents trained on reference answers at matched compute ...”

Shorter; preserves the useful comparison without asserting that a finite training run is a mathematical upper bound. The proposed “half its gain” criterion should be described as an empirical target, not a guarantee.

C34. Concrete cut ledger: recover space without deleting the mathematical case

Location: all technical sections, pp. 1-24. **Priority after correcting C5-C7's claims.**

Cut/reduction	What to preserve	Expected role in the page budget
Delete Figure 1, Table 1 and Figure 7 plus their callouts	Figure 2 running example; Figure 3 measured simulation; Figure 5 roles; Figure 6 harness	Recover redundant graphic areas. Table 1 also avoids the brittle “no current approach” classification.
Shorten Section 1's repeated motivation, scale claims and broad use-case list by about 200-300 words	Central question, four gaps, named focus area, one naval consequence	Recover prose; use C9/C28 rather than adding a new motivation.
Shorten the three RG Challenges/Overview passages by about 350-450 words collectively	One explicit failure-to-task mapping per RG	Their failures already appear on pp. 1-3; preserve the task methods instead.
Shorten Section 2's historical inventories by about 200-300 words	Closest model/debate/peer-reward predecessors and the precise gap	Preserve state-of-the-art awareness without a catalogue and duplicate table.
Cut about 350-450 words of repeated post-equation explanations, simulation retelling, generic harness description and tax analogy	Conditional assumptions, actual judge training, audit research target	Familiar derivations can remain concise; do not delete the new quality argument.
Shorten duplicated achievement/service and generic support assertions by about 250-350 words	Qualifications tied directly to the tasks; all responsibility topics; truthful support boundary	Sections 4-6 must remain inside 20 pages.
Compress duplicate activity descriptions across Tables 3-4 by about 150-200 words	Dates, milestones, measurable primary outcomes	Avoid saying every task's method a third time in the schedule and goals table.

These ranges target roughly 1,500-2,050 prose words, plus redundant figures/table areas and any unrun-pilot savings. Depending on font decisions, further cuts may be necessary. They are a planning ledger; only the final compiled PDF can establish compliance. Shorten Figure 2 and Figure 6 labels to actions and evidence before increasing their font size; do not enlarge every existing paragraph inside a box unchanged. Do not spend the same cut twice on both new prose and font expansion.

Literature checks and final priorities

C35. What the novelty/source check establishes, and what it does not

Location: Section 2 pp. 5-6; mathematical rationale pp. 9-19.

Primary-source checks support the following boundaries. These are sources for the review, not a request for another bibliography paragraph:

- Jaffe et al. (2016), Unsupervised Ensemble Learning with Dependent Classifiers: the model hierarchy is a close predecessor, now acknowledged directly. The new work must lie in routing, learning groups under specified failures and strategic use, not the existence of latent dependence alone.
- Balasubramanian et al. (2026), dependence-aware Ising aggregation and Zhao et al. (2026), CARE: dependence-aware LLM aggregation and associated theory exist. Generic “shared mistakes were not addressed” claims remain too broad; your particular debate/strategic problem can still be new.
- Jeong et al. (2026), Tool-MAD and Li et al. (2026), DynaDebate: tool-assisted debate and process critique are antecedents. The proposed exact-call requirement, impartial selection and audits should be the comparison dimensions. I did not establish exhaustive priority for replay.
- Qiu et al. (2026), Truthfulness Despite Weak Supervision and Yang et al. (2026), Co-RL: peer-derived training already exists. Qiu's appendix studies collusive composition and describes limitations; distinguish lacking a guarantee for your threat model from ignoring collusion. These remain relevant future comparisons, not new deadline experiments.
- Natarajan et al. (2013), Learning with Noisy Labels: noisy-label theory is an appropriate lineage, but it does not prove message-credit alignment or arbitrary label-free GRPO improvement.
- Howard et al. (2021), Time-Uniform, Nonparametric, Nonasymptotic Confidence Sequences: supplies statistical tools; the role-specific online harm result is a proposed contribution, not supplied by citation.
- Khan et al. (2024), Debating with More Persuasive LLMs Leads to More Truthful Answers, Appendix A.5: supports the stated choice to fine-tune a judge on correct labels rather than imitate human reasoning in that setup. Keep the concrete training rationale.

What to do now: first resolve 24-page compliance and the remaining pending evidence; then make the C5-C7 claim corrections and C8 evaluation distinction; then apply the small naval/support substitutions. Do not reopen RG1's architecture or pursue minor theorem edge cases at the expense of those fixes. The hard mathematical message in v2 is narrower than v1: most component arguments now make sense, but the learning and audit claims must stop where their assumptions stop.